Unit: Unit 8: Quadratics **Name:** ____________

Topic: Graphing Quadratics and Properties of Parabolas **Date:** __________

Foundation (Jalapeno)

Q1. What is the vertex of the parabola $y = x^2 + 4x + 4$?
(A) $(-2, 0)$
(B) $(-2, 4)$
(C) $(0, 4)$
(D) $(2, 0)$
(E) $(0, 0)$

Q2. Which axis of symmetry corresponds to the parabola $y = x^2 - 7x + 10$?
(A) $x = -3$
(B) $x = 3.5$
(C) $x = 0$
(D) $x = 7$
(E) $y = 0$

Q3. What are the x-intercepts of the parabola $y = x^2 + 2x - 3$?
(A) $(-3, 0)$, $(1, 0)$
(B) $(-1, 0)$, $(3, 0)$
(C) $(-2, 0)$, $(1.5, 0)$
(D) $(0, 0)$, $(3, 0)$
(E) $(-1.5, 0)$, $(2, 0)$

Q4. What is the direction of the parabola $y = -x^2 + 1$?
(A) Opens upwards
(B) Opens downwards
(C) Opens to the left
(D) Opens to the right
(E) No direction

Q5. What is the maximum value of the function $y = -x^2 + 4x - 3$?
(A) -1
(B) 1
(C) 4
(D) 5
(E) 6

Q6. Which parabola has the equation $y = (x - 2)^2 - 1$?
(A) Vertex at $(0, -1)$
(B) Vertex at $(2, -1)$
(C) Vertex at $(2, 1)$
(D) Vertex at $(-2, 1)$
(E) Vertex at $(1, 0)$

Q7. What are the coordinates of the vertex of the parabola $y = 2x^2 - 4x - 3$?
(A) $(1, -5)$
(B) $(0, -3)$
(C) $(-1, 0)$
(D) $(0.5, -3.5)$
(E) $(2, 1)$

Q8. What is the y-intercept of the parabola $y = 2x^2 - 5x - 3$?
(A) $(0, 2)$
(B) $(0, -1)$
(C) $(0, -3)$
(D) $(0, 3)$
(E) $(0, 5)$

Open Ended Questions (FRQ)

Q9. Graph the parabola $y = x^2 + 2x - 6$ and identify its vertex, axis of symmetry, x-intercepts, y-intercept, and direction. (Show your work and use proper labeling.)

Q10. Find the equation of a parabola with vertex $(1, 2)$, y-intercept $(0, 0)$, and axis of symmetry parallel to the y-axis.

Chef's Tips

- Tip 1: Focus on identifying key features such as the vertex, axis of symmetry, and intercepts when graphing parabolas.
- Tip 2: Use the vertex form $y = a(x - h)^2 + k$ to easily identify the vertex (h, k).
- Tip 3: Determine the direction of the parabola by analyzing the coefficient of x^2.